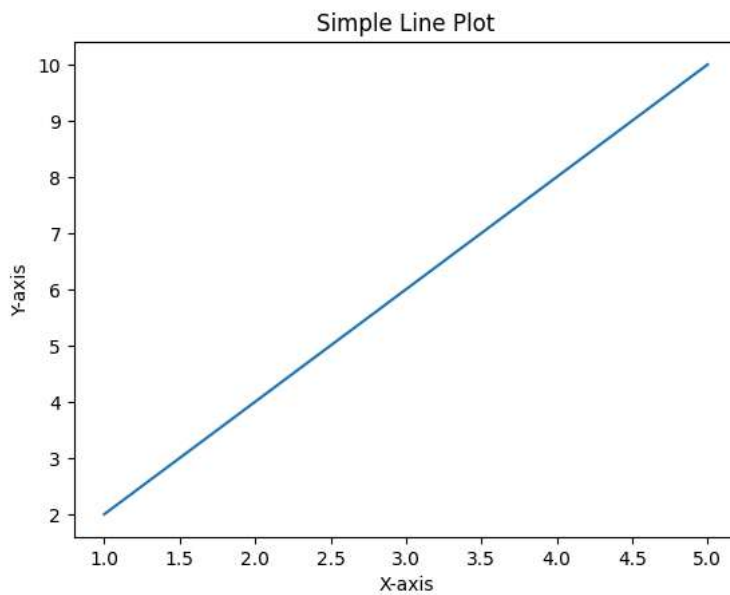


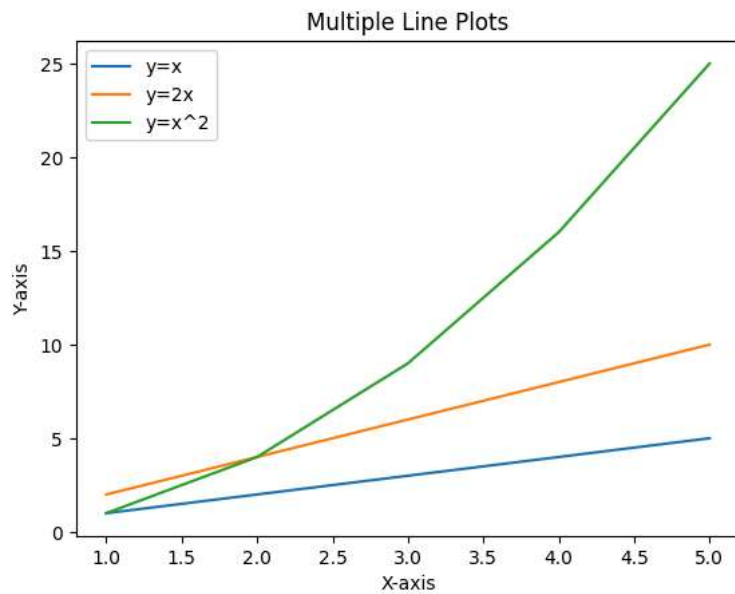
```
Name="BALANAGALAKSHMI B"  
Reg_No="111924AI01010"  
print("Name:",Name)  
print("Reg No:",Reg_No)
```

Name: BALANAGALAKSHMI B
Reg No: 111924AI01010

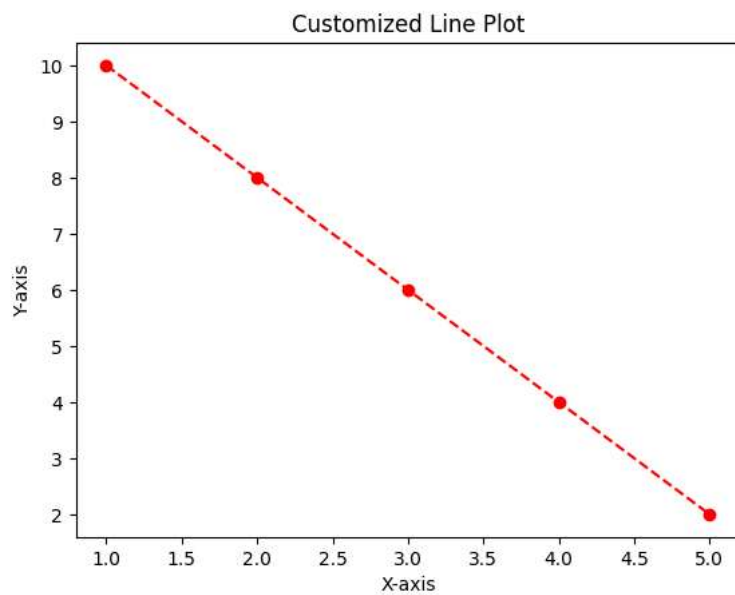
```
#Simple Line Plot  
import matplotlib.pyplot as plt  
#data  
X=[1,2,3,4,5]  
Y=[2,4,6,8,10]  
plt.plot(X,Y)  
#Add labels  
plt.xlabel("X-axis")  
plt.ylabel("Y-axis")  
plt.title("Simple Line Plot")  
plt.show()
```



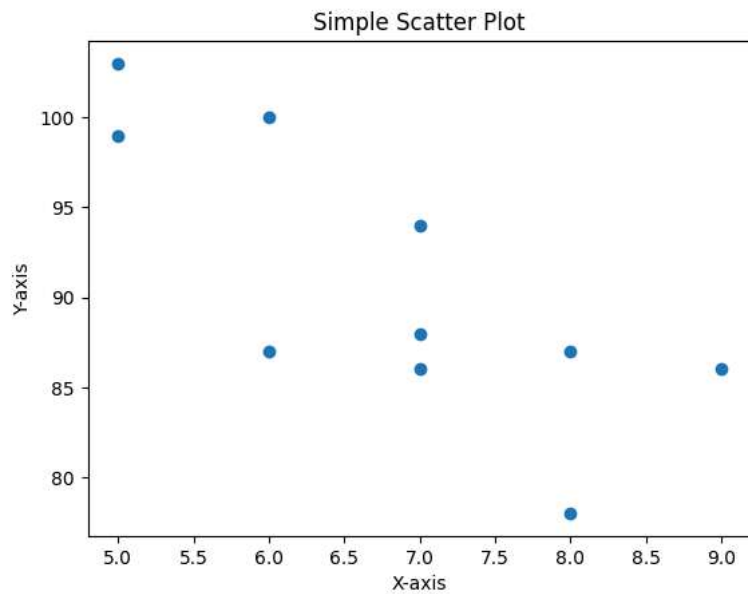
```
#Multiple Lines  
import matplotlib.pyplot as plt  
X=[1,2,3,4,5]  
Y1=[1,2,3,4,5]  
Y2=[2,4,6,8,10]  
Y3=[1,4,9,16,25]  
plt.plot(X,Y1,label="y=x")#line 1  
plt.plot(X,Y2,label="y=2x")#line 2  
plt.plot(X,Y3,label="y=x^2")#line 3  
plt.xlabel("X-axis")  
plt.ylabel("Y-axis")  
plt.title("Multiple Line Plots")  
plt.legend()  
plt.show()
```



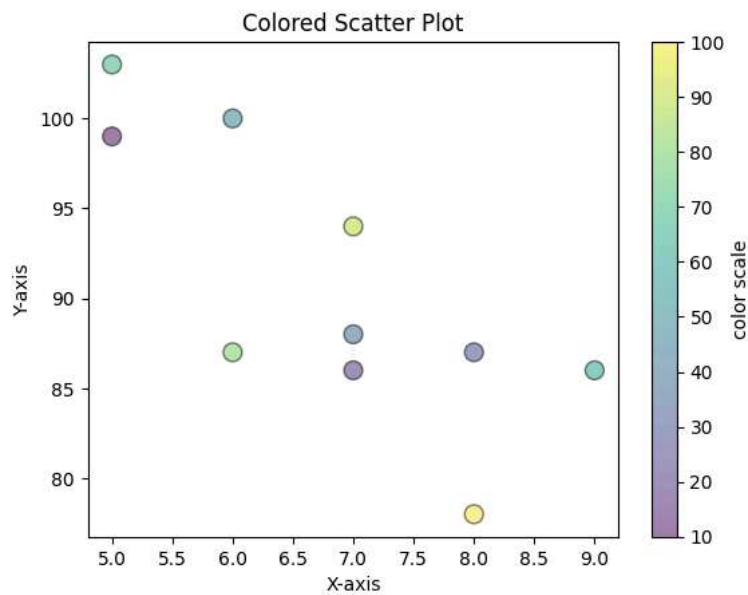
```
#Customized Line
import matplotlib.pyplot as plt
X=[1,2,3,4,5]
Y=[10,8,6,4,2]
plt.plot(X,Y,color='red',linestyle='--',marker='o')
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Customized Line Plot")
plt.show()
```



```
#Scatter Plots in Matplotlib
#1.Simple Scatter Plot
import matplotlib.pyplot as plt
#data
X=[5,7,8,7,6,9,5,6,7,8]
Y=[99,86,87,88,100,86,103,87,94,78]
plt.scatter(X,Y)
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Simple Scatter Plot")
plt.show()
```

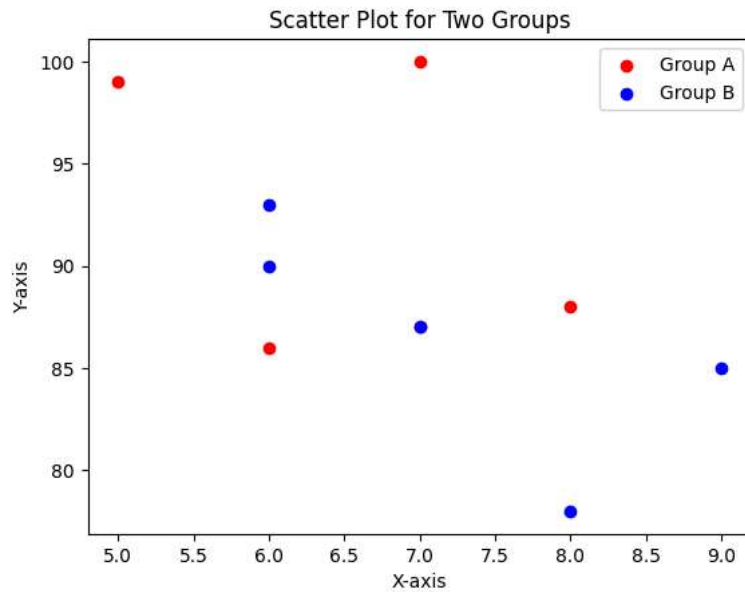


```
#2.Colored Scatter Plot
import matplotlib.pyplot as plt
X=[5,7,8,7,6,9,5,6,7,8]
Y=[99,86,87,88,100,86,103,87,94,78]
colors=[10,20,30,40,50,60,70,80,90,100]
plt.scatter(X,Y,c=colors,cmap='viridis',s=100,alpha=0.5,edgecolors='black')
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title('Colored Scatter Plot')
plt.colorbar(label="color scale")
plt.show()
```

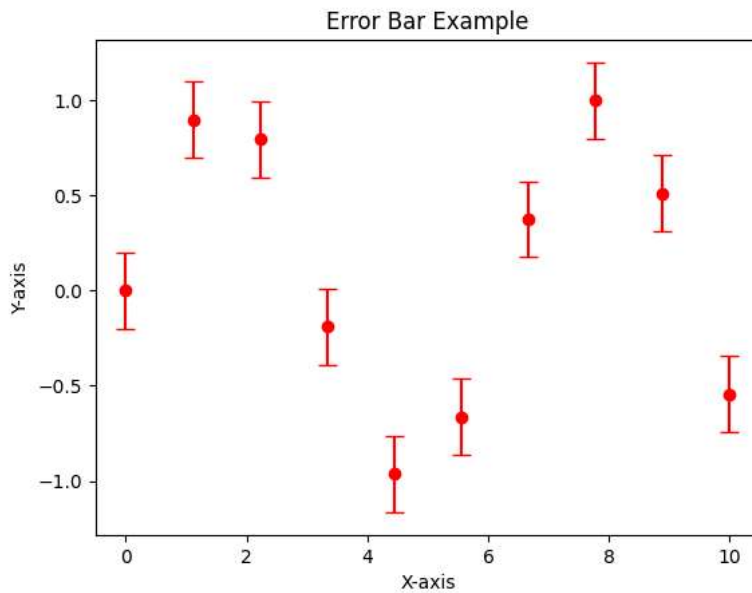


```
#3.Comparing two groups
import matplotlib.pyplot as plt
#Group A
X1=[5,6,7,8,7]
Y1=[99,86,87,88,100]
#Group B
X2=[6,7,8,9,6]
Y2=[93,87,78,85,90]
plt.scatter(X1,Y1,color='red',label="Group A")
plt.scatter(X2,Y2,color='blue',label="Group B")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Scatter Plot for Two Groups")
```

```
plt.legend()
plt.show()
```

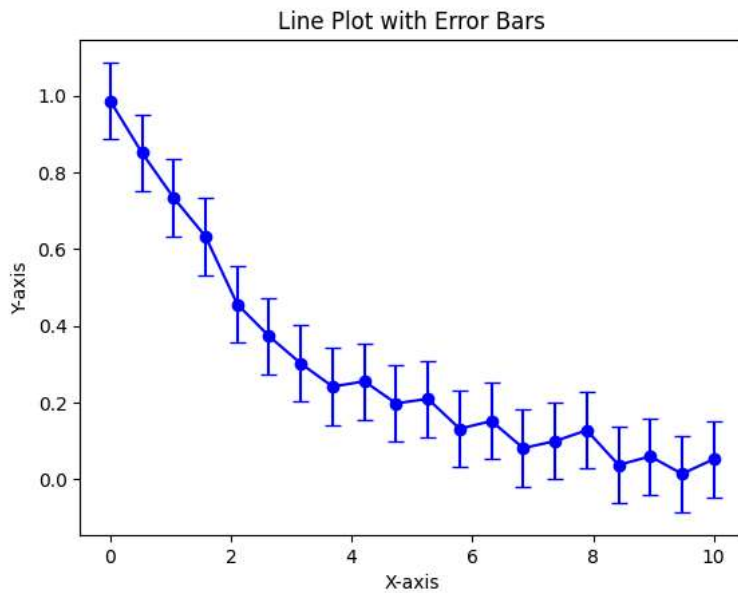


```
#Visualizing Errors in Matplotlib
#1.Simple Error Bars
import matplotlib.pyplot as plt
import numpy as np
X=np.linspace(0,10,10)
Y=np.sin(X)
#Assume errors in Y values
errors=0.2
plt.errorbar(X,Y,yerr=errors,fmt='o',capsize=5,color='red')
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Error Bar Example")
plt.show()
```

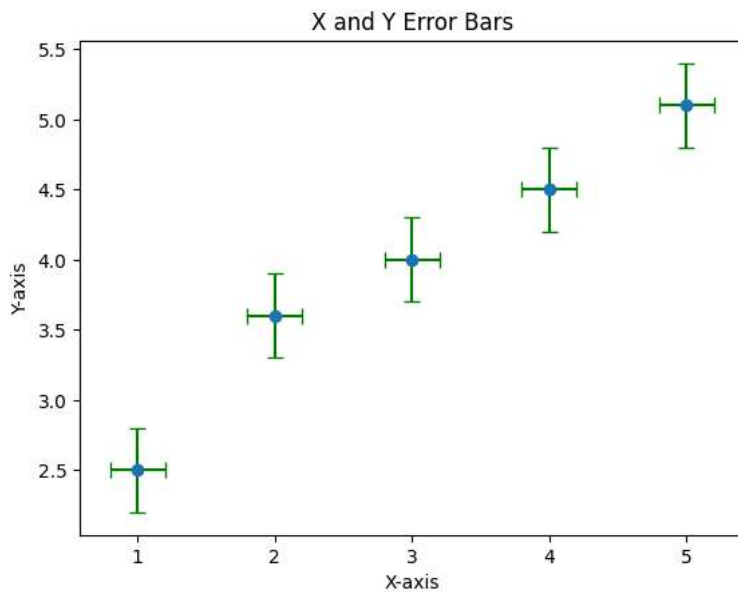


```
#2.With Line and Error Bars
import matplotlib.pyplot as plt
import numpy as np
X=np.linspace(0,10,20)
Y=np.exp(-X/3)+np.random.normal(0,0.05,size=20)
plt.errorbar(X,Y,yerr=0.1,fmt='-o',color='blue',capsize=4)
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
```

```
plt.title("Line Plot with Error Bars")
plt.show()
```

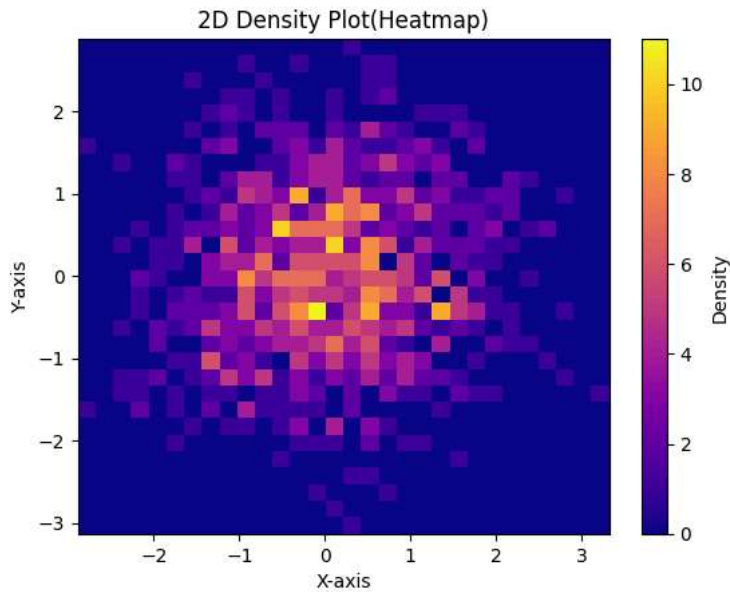


```
#3.Both X and Y Errors
import matplotlib.pyplot as plt
import numpy as np
X=np.array([1,2,3,4,5])
Y=np.array([2.5,3.6,4.0,4.5,5.1])
X_error=0.2
Y_error=0.3
plt.errorbar(X,Y,xerr=X_error,yerr=Y_error,fmt='o',ecolor='green',capsize=4)
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("X and Y Error Bars")
plt.show()
```

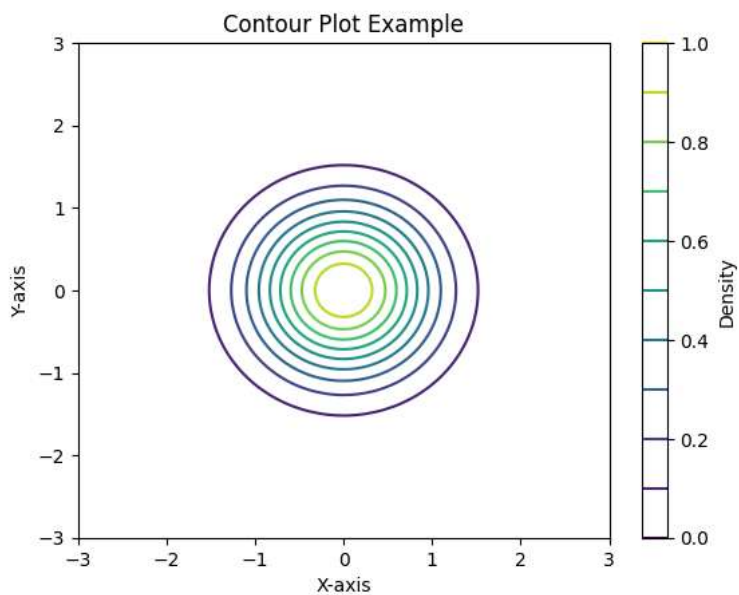


```
#Density Plot
import numpy as np
import matplotlib.pyplot as plt
#Generate random 2D data
X=np.random.randn(1000)
Y=np.random.randn(1000)
#2D Histogram (density heatmap)
plt.hist2d(X,Y,bins=30,cmap='plasma')
plt.colorbar(label="Density")
plt.xlabel("X-axis")
```

```
plt.ylabel("Y-axis")
plt.title("2D Density Plot(Heatmap)")
plt.show()
```

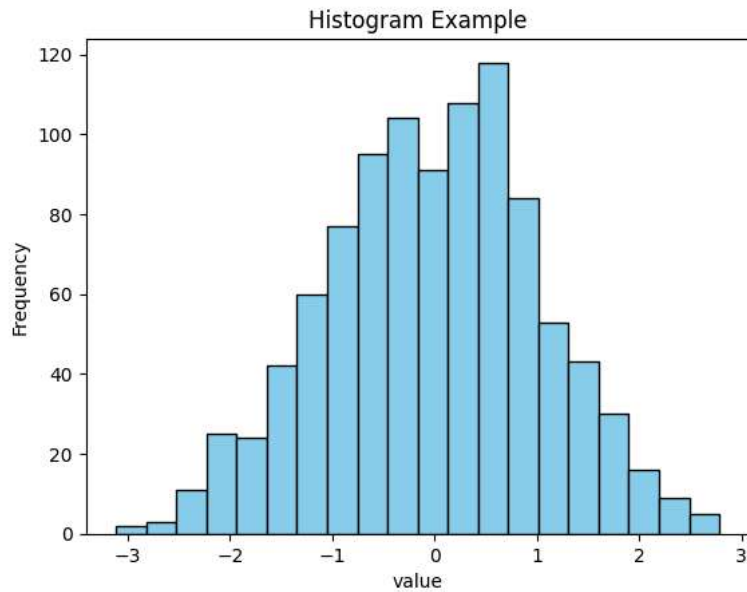


```
#Contour Plot
import numpy as np
import matplotlib.pyplot as plt
#create grid of X&Y values
X=np.linspace(-3,3,100)
Y=np.linspace(-3,3,100)
X,Y=np.meshgrid(X,Y)
#Define a 2D function
Z=np.exp(-(X**2+Y**2))
plt.contour(X,Y,Z,levels=10,cmap='viridis')
plt.colorbar(label="Density")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Contour Plot Example")
plt.show()
```



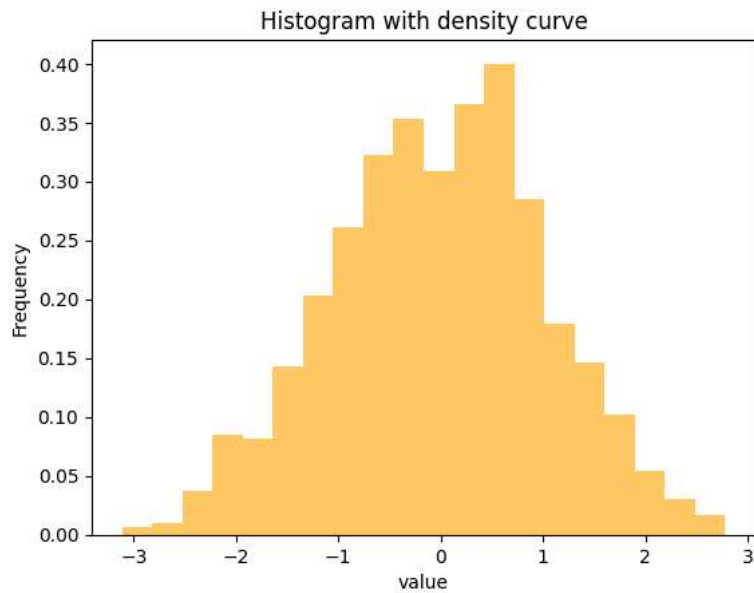
```
#Basic Histogram
import numpy as np
import matplotlib.pyplot as plt
data=np.random.randn(1000)
plt.hist(data,bins=20,color='skyblue',edgecolor='black')
plt.xlabel("value")
```

```
plt.ylabel("Frequency")
plt.title("Histogram Example")
plt.show()
```

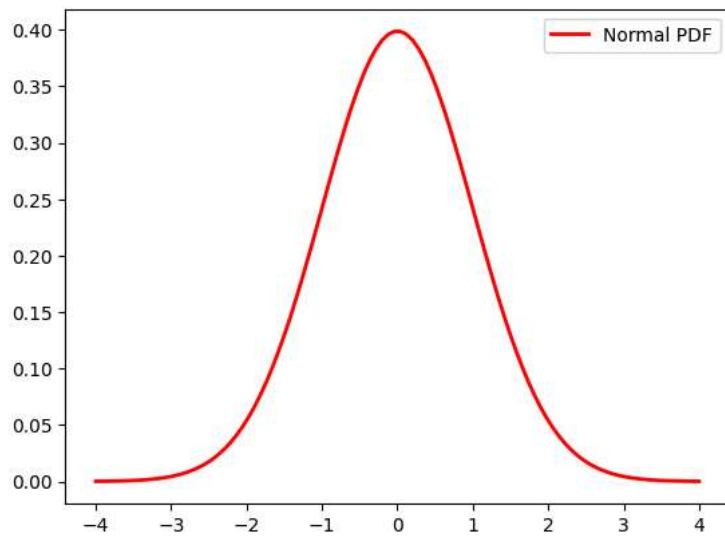


```
#Histogram with density curve
plt.hist(data,bins=20,density=True,alpha=0.6,color='orange')
plt.title("Histogram with density curve")
plt.xlabel("value")
plt.ylabel("Frequency")
```

```
Text(0, 0.5, 'Frequency')
```



```
#overlay density line
from scipy.stats import norm
X=np.linspace(-4,4,100)
plt.plot(X,norm.pdf(X,0,1),'r-',lw=2,label='Normal PDF')
plt.legend()
plt.show()
```



```
#Binning
import matplotlib.pyplot as plt
import numpy as np
data=np.random.randn(1000)
plt.hist(data,bins=[-3,-2,-1,0,1,2,3],color='skyblue',edgecolor='black')
plt.xlabel("value")
plt.ylabel("Frequency")
plt.title("Histogram with 5 Bins")
plt.show()
```

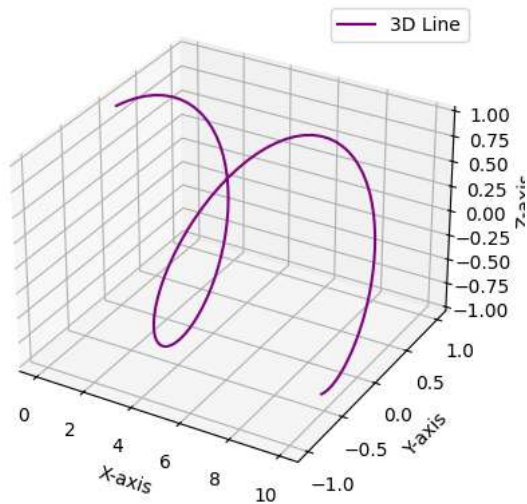
[Show hidden output](#)

```
#3D Plotting in Matplotlib
#Import the 3D Toolkit
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt
import numpy as np
#create a 3D Axes Object
fig=plt.figure()
ax=fig.add_subplot(111,projection='3d')
```

[Show hidden output](#)

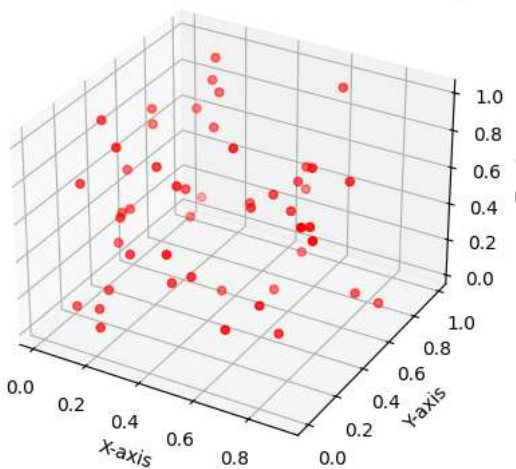
```
#1.3D Line Plot
X=np.linspace(0,10,100)
Y=np.sin(X)
Z=np.cos(X)
fig=plt.figure()
ax=fig.add_subplot(111,projection='3d')
ax.plot(X,Y,Z,color='purple',label='3D Line')
ax.set_xlabel('X-axis')
ax.set_ylabel('Y-axis')
ax.set_zlabel('Z-axis')
ax.legend()
plt.title("3D Line Plot Example")
plt.show()
```


3D Line Plot Example



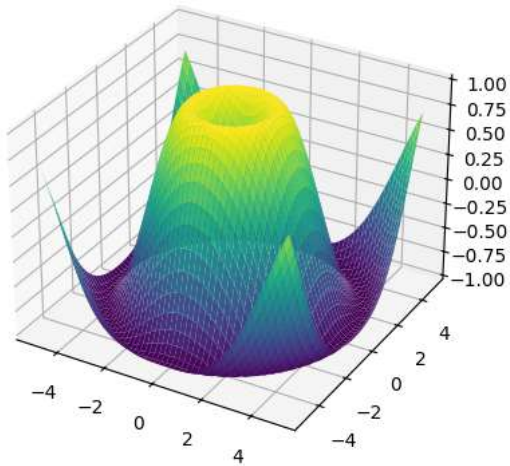
```
#2.3D Scatter Plot
X=np.random.rand(50)
Y=np.random.rand(50)
Z=np.random.rand(50)
fig=plt.figure()
ax=fig.add_subplot(111,projection='3d')
ax.scatter(X,Y,Z,color='red',marker='o')
ax.set_xlabel("X-axis")
ax.set_ylabel("Y-axis")
ax.set_zlabel("Z-axis")
ax.legend()
```

/tmp/ipython-input-2480577815.py:11: UserWarning: No artists with labels found to put in legend. Note that artists whose label sta
ax.legend()
<matplotlib.legend.Legend at 0x7f35d13f4fe0>



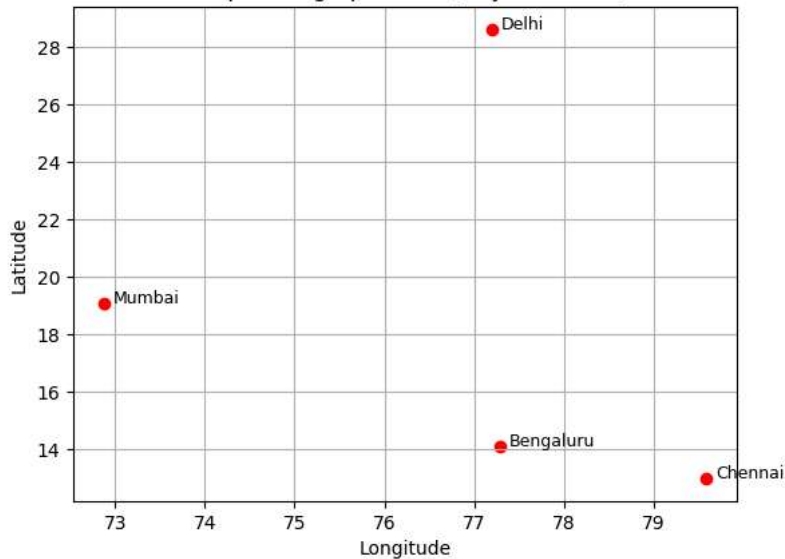
```
#3.3D Surface Plot
X=np.linspace(-5,5,50)
Y=np.linspace(-5,5,50)
X,Y=np.meshgrid(X,Y)
Z=np.sin(np.sqrt(X**2+Y**2))
fig=plt.figure()
ax=fig.add_subplot(111,projection='3d')
ax.plot_surface(X,Y,Z,cmap='viridis')
ax.set_title("3D Surface Plot")
plt.show()
```

3D Surface Plot



```
#Geographic Data Visualization
#1.Simple Latitude/Longitude Plot
import matplotlib.pyplot as plt
latitude=[12.97,28.61,19.07,14.08]
longitude=[79.59,77.20,72.87,77.29]
cities=['Chennai','Delhi','Mumbai','Bengaluru']
plt.scatter(longitude,latitude,color='red',marker='o')
for i,city in enumerate(cities):
    plt.text(longitude[i]+0.1,latitude[i],city,fontsize=9)
plt.xlabel('Longitude')
plt.ylabel('Latitude')
plt.title("Simple Geographic Plot (City Locations)")
plt.grid(True)
plt.show()
```

Simple Geographic Plot (City Locations)



```
#2.Using Cartopy (for real maps)
import cartopy.crs as ccrs
import matplotlib.pyplot as plt
plt.figure(figsize=(6,6))
ax=plt.axes(projection=ccrs.PlateCarree())
ax.coastlines()
ax.set_extent([68,98,8,38])
latitude = [12.97,28.61,19.07,14.08]
longitude = [79.59,77.20,72.87,77.29]
```

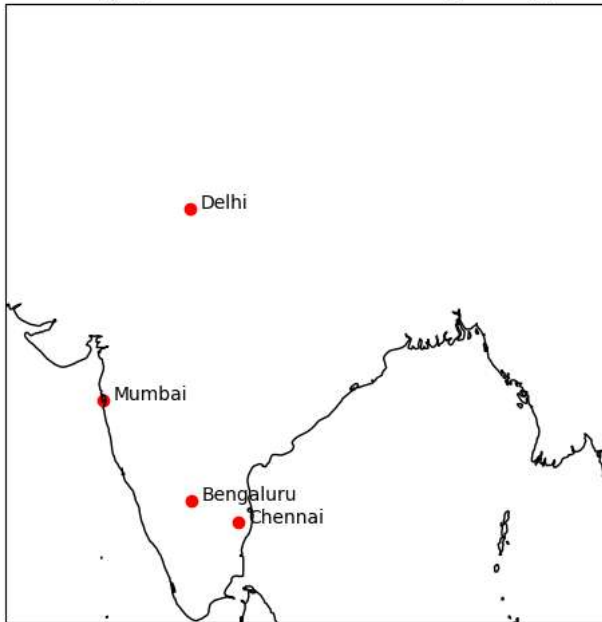
```

cities = ['Chennai','Delhi','Mumbai','Bengaluru']
ax.scatter(longitude,latitude,color='red',transform=ccrs.PlateCarree())
for i,city in enumerate(cities):
    plt.text(longitude[i]+0.5,latitude[i],city,transform=ccrs.PlateCarree())
plt.title("Geographic Data Visualization Using Cartopy")
plt.show()

```

/usr/local/lib/python3.12/dist-packages/cartopy/io/__init__.py:242: DownloadWarning: Downloading: <https://naturalearth.s3.amazonaws.com>
warnings.warn(f'Downloading: {url}', DownloadWarning)

Geographic Data Visualization Using Cartopy



```

#Statsmodels
#Simple Linear Regression
import statsmodels.api as sm
import pandas as pd
data=pd.DataFrame({'Hours':[1,2,3,4,5], 'Scores':[25,47,52,73,88]})
df=pd.DataFrame(data)
print(df)

```

	Hours	Scores
0	1	25
1	2	47
2	3	52
3	4	73
4	5	88

```

#Define X&Y
X=data['Hours']
Y=data['Scores']
#Add constant to X (for intercept)
X=sm.add_constant(X)
#Build model
model=sm.OLS(Y,X).fit()
#show summary
print(model.summary())

```

```

=====
                        OLS Regression Results
=====
Dep. Variable:          Scores   R-squared:                0.977
Model:                  OLS     Adj. R-squared:            0.969
Method:                 Least Squares   F-statistic:            124.7
Date:                   Sat, 11 Oct 2025   Prob (F-statistic):      0.00154
Time:                   04:49:03   Log-Likelihood:         -13.117
No. Observations:        5         AIC:                   30.23
Df Residuals:            3         BIC:                   29.45
Df Model:                1
Covariance Type:        nonrobust
=====
                        coef    std err          t      P>|t|    [0.025    0.975]
=====

```

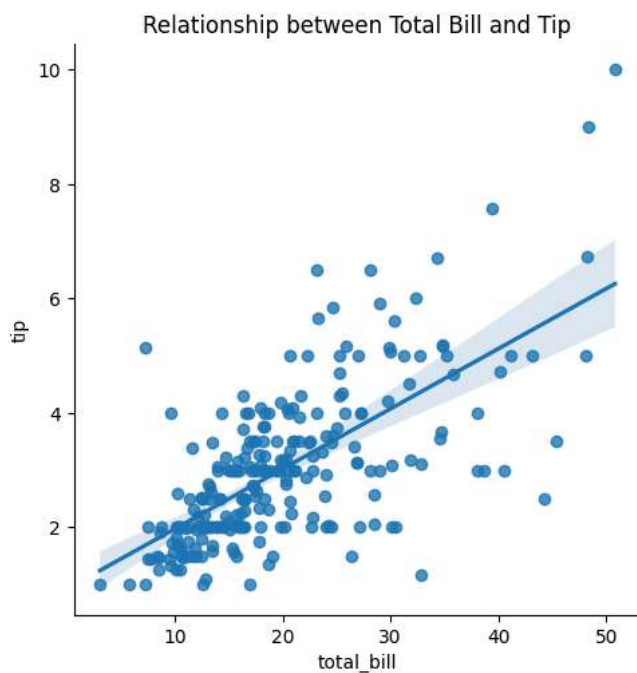
const	11.4000	4.515	2.525	0.086	-2.969	25.769
Hours	15.2000	1.361	11.165	0.002	10.868	19.532
=====						
Omnibus:		nan	Durbin-Watson:			3.309
Prob(Omnibus):		nan	Jarque-Bera (JB):			0.135
Skew:		0.066	Prob(JB):			0.935
Kurtosis:		2.205	Cond. No.			8.37
=====						

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

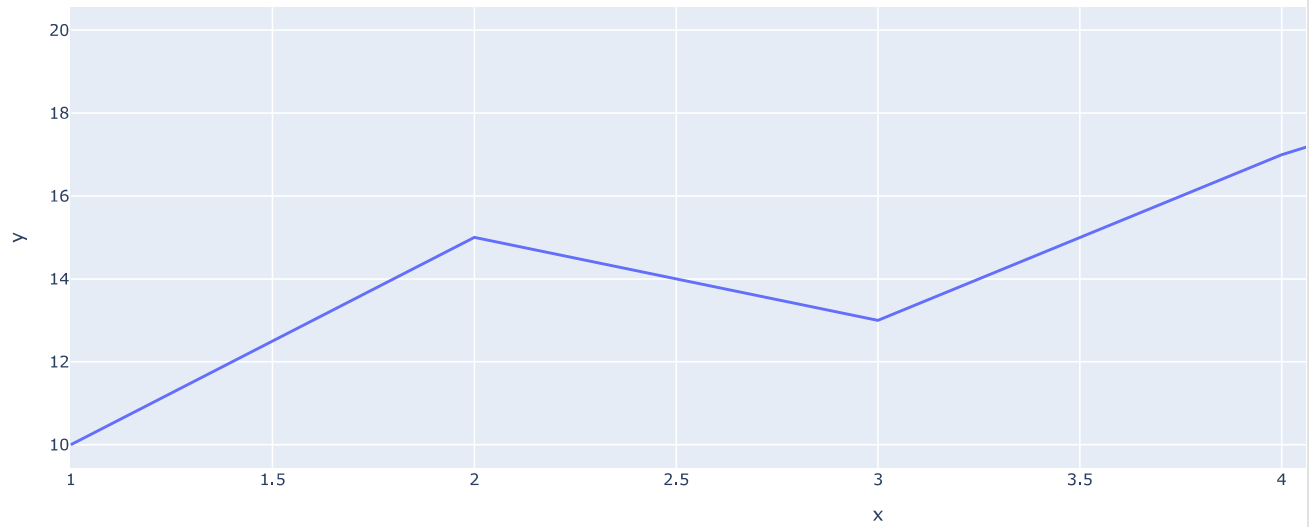
/usr/local/lib/python3.12/dist-packages/statsmodels/stats/stattools.py:74: ValueWarning: omni_normtest is not valid with less than 8 observations; %i "

```
#Seaborn
#Linear Relationship Plot
import seaborn as sns
import matplotlib.pyplot as plt
tips=sns.load_dataset("tips")
sns.lmplot(x="total_bill",y="tip",data=tips)
plt.title("Relationship between Total Bill and Tip")
plt.show()
```



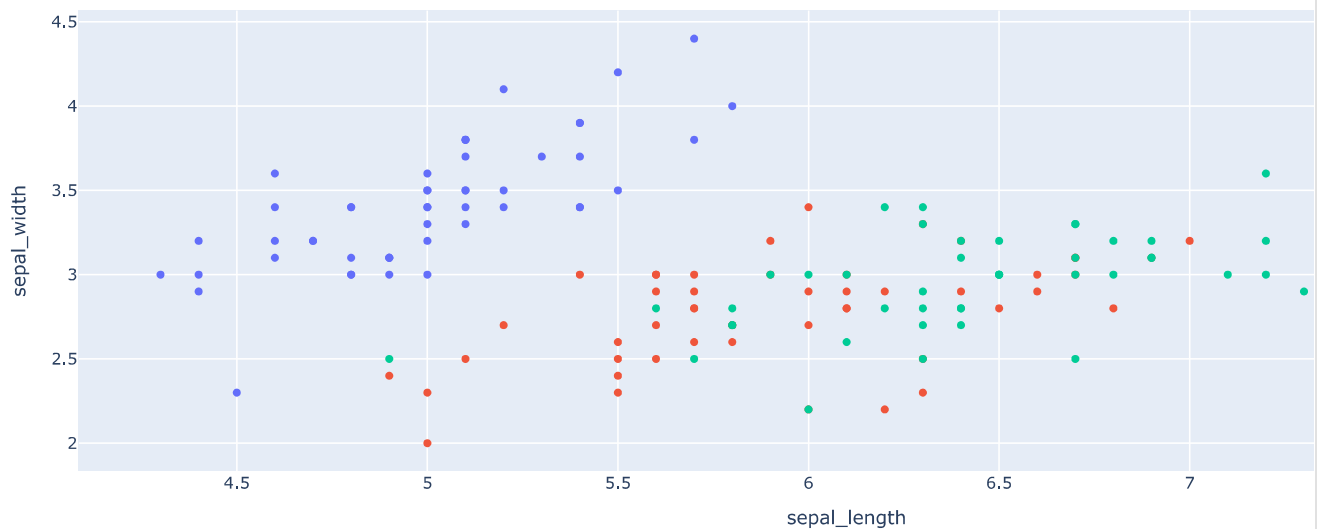
```
#Graph Plotting Using Plotly
#1.Basic Line Plot
import plotly.express as px
data={'x':[1,2,3,4,5],
      'y':[10,15,13,17,20]}
fig=px.line(data,x='x',y='y',title='Simple Line Plot')
fig.show()
```

Simple Line Plot



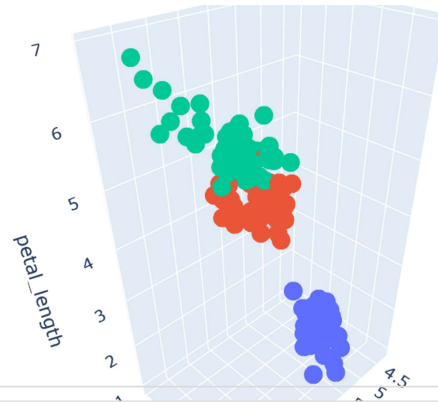
```
#2.Scatter Plot
import plotly.express as px
data=px.data.iris()
fig=px.scatter(data,x="sepal_length",y="sepal_width",color="species",title="IrisDataset-sepal width Vs sepal length")
fig.show()
```

IrisDataset-sepal width Vs sepal length



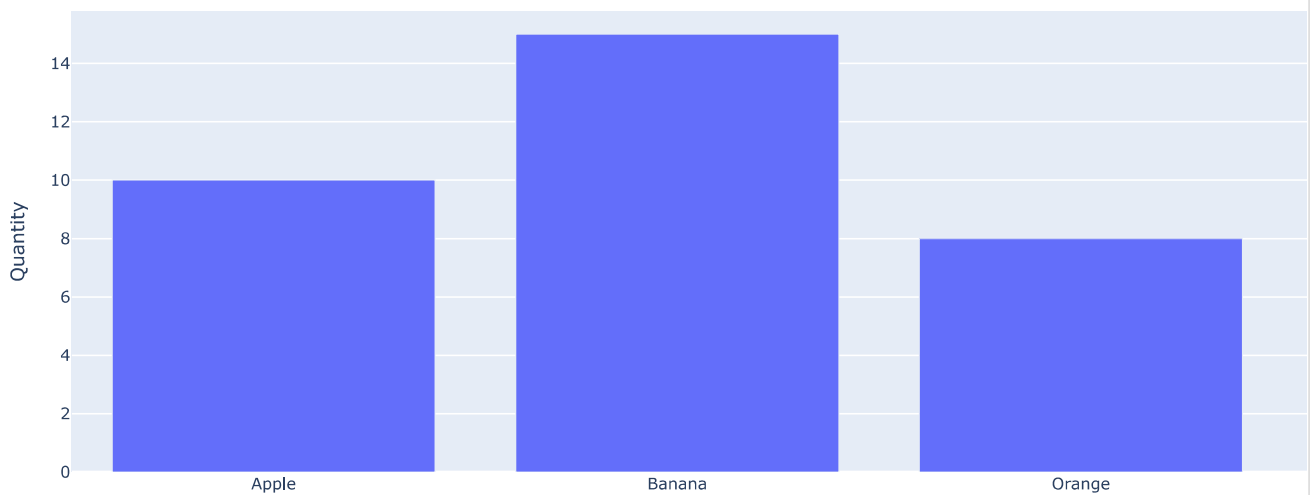
```
#3.Scatter 3d Plot
import plotly.express as px
fig=px.scatter_3d(px.data.iris(),x='sepal_length',y='sepal_width',z='petal_length',color='species',title='3D Scatter Plot-IrisData')
fig.show()
```

3D Scatter Plot-IrisDataset



```
#4. Bar Plot
import plotly.express as px
data={'Fruits':['Apple','Banana','Orange','Grapes'],
      'Quantity':[10,15,8,12]}
fig=px.bar(data,x='Fruits',y='Quantity',title='Fruit Sales')
fig.show()
```

Fruit Sales



```
# BOKEH
from bokeh.plotting import figure
from bokeh.io import show, output_notebook
output_notebook()
#create figure
p=figure(title="interactive scatter plot", x_axis_label='x',y_axis_label='y', tools='pan, wheel_zoom, box_zoom, reset, hover')
#add data
x= [1,2,3,4,5]
y=[6,7,2,4,5]
p.scatter(x, y, size=10, color='navy', alpha=0.5)
show(p)
```

interactive scatter plot

